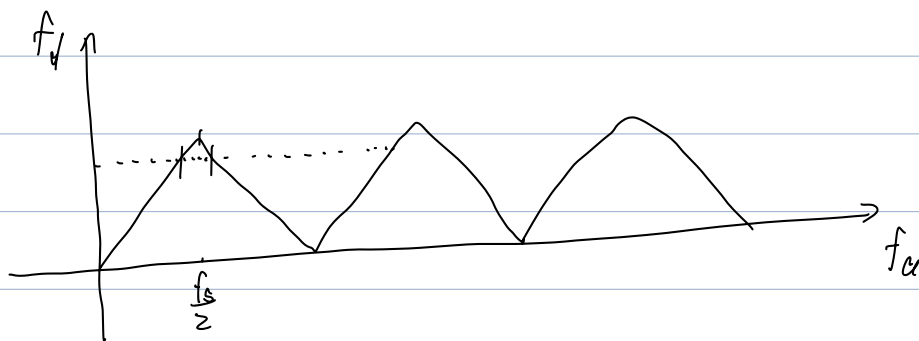




$$f_s = 200 \text{ MHz}$$

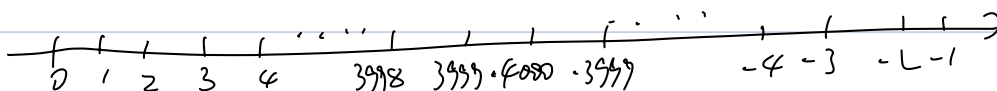
$$f_N = 100 \text{ MHz}$$

Q1. (4). $f_s = 8 \text{ kHz}$, $T = 1 \text{ s} \Rightarrow N = f_s \cdot T = 8000$

DFT bin spacing $\Delta f = \frac{f_s}{N} = 1 \text{ Hz}$.

pos freq: $0 \sim 100 \text{ Hz} \rightarrow k: 0 \sim 100$

neg freq: $-100 \sim 0 \text{ Hz} \rightarrow k: 7900 \sim 8000$



$$Y(k) = \begin{cases} 0 & \dots \\ X(k) & \dots \end{cases}$$

$$\Delta f = \frac{f_s}{N} \quad k_c = \frac{f_c}{\Delta f} = \frac{0.5}{100/N} = \frac{N}{1000}$$

$\Delta f = 0.5 \text{ Hz}$. $\frac{f_s}{\Delta f} = \frac{1000}{0.5} = 1000 = N$ at least 1000.